Database System Engineering and Distributed Backend Development

Course Code: 25CS1302E

DDL, DML, DCL, TCL Commands

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Section - 3

Create the database:

```
CREATE DATABASE IF NOT EXISTS bookflow1_db;
```

Output:

```
88  15:52:08  CREATE DATABASE IF NOT EXISTS bookflow1_db
```

Use database:

```
USE bookflow1_db;
```

Output:

```
89  15:53:05  USE bookflow1_db
```

DDL (DATA DEFINITION LANGUAGE)

Create table:

```
CREATE TABLE bank_transactions (
    txn_id INT PRIMARY KEY,
    customer_name VARCHAR(50),
    branch_name VARCHAR(50),
    transaction_type VARCHAR(20),
    amount DECIMAL(10,2),
    transaction_date DATE
);
SELECT * FROM bank_transactions;
```

Output:

	txn_id	customer_name	branch_name	transaction_type	amount	transaction_date
*	NULL	NULL	NULL	NULL	NULL	NULL

Alter table add column:

```
ALTER TABLE bank_transactions
ADD account_no VARCHAR(20);
SELECT * FROM bank_transactions;
```

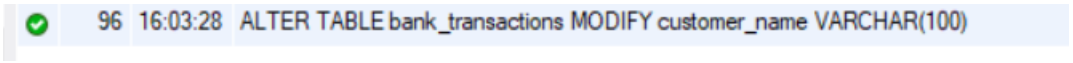
Output:

	txn_id	customer_name	branch_name	transaction_type	amount	transaction_date	account_no
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Alter table modify column

```
ALTER TABLE bank_transactions
MODIFY customer_name VARCHAR(100);
```

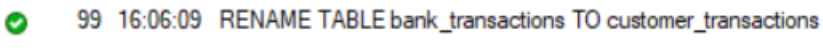
Output:



Rename table

```
RENAME TABLE bank_transactions
TO customer_transactions;
```

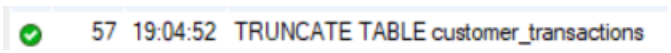
Output:



Truncate table:

```
TRUNCATE TABLE customer_transactions;
```

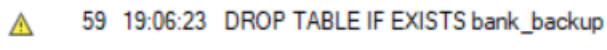
Output:



Drop table

```
DROP TABLE IF EXISTS bank_backup;
```

Output:



DML (DATA MANIPULATION LANGUAGE)

Create table:

```sql
CREATE TABLE bank_transactions (
    txn_id INT PRIMARY KEY,
    customer_name VARCHAR(50),
    branch_name VARCHAR(50),
    transaction_type VARCHAR(20),
    amount DECIMAL(10,2),
    transaction_date DATE
);
```

Output:

	txn_id	customer_name	branch_name	transaction_type	amount	transaction_date
*	NULL	NULL	NULL	NULL	NULL	NULL

Insert sample data:

```sql
INSERT INTO bank_transactions VALUES
(101,'Sushma','Hyderabad','Deposit',5000,'2024-01-05'),
(102,'Sowmya','Hyderabad','Withdrawal',2000,'2024-01-06'),
(103,'Mokshagna','Vijayawada','Deposit',12000,'2024-01-08'),
(104,'Venugopal','Vizag','Deposit',8000,'2024-01-10'),
(105,'Kalavathi','Hyderabad','Withdrawal',3500,'2024-01-11'),
(106,'Govinda','Vizag','Deposit',15000,'2024-01-12'),
(107,'Krishna','Vijayawada','Withdrawal',1000,'2024-01-13'),
(108,'Sairam','Hyderabad','Deposit',9000,'2024-01-14'),
(109,'Laxmi','Vizag','Withdrawal',4000,'2024-01-15'),
(110,'Saraswati','Vijayawada','Deposit',11000,'2024-01-16');
SELECT * FROM bank_transactions;
```

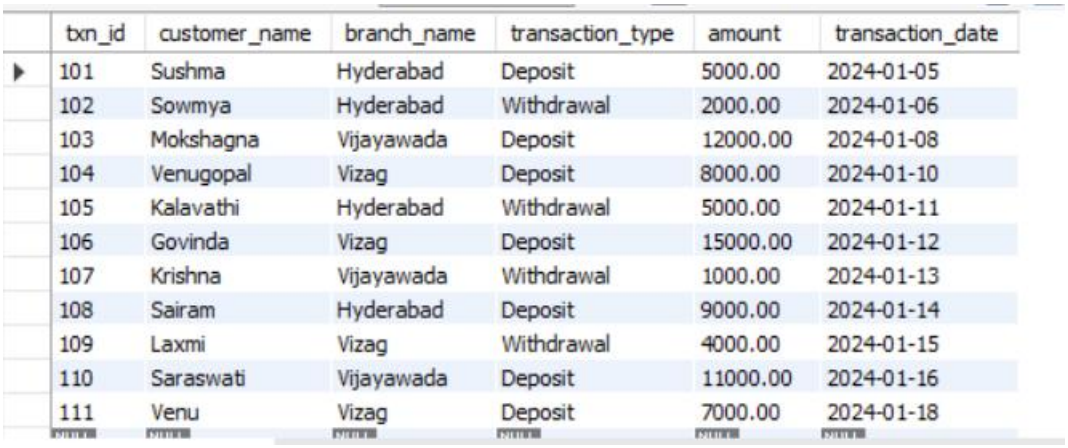
Output:

	txn_id	customer_name	branch_name	transaction_type	amount	transaction_dat
▶	101	Sushma	Hyderabad	Deposit	5000.00	2024-01-05
	102	Sowmya	Hyderabad	Withdrawal	2000.00	2024-01-06
	103	Mokshagna	Vijayawada	Deposit	12000.00	2024-01-08
	104	Venugopal	Vizag	Deposit	8000.00	2024-01-10
	105	Kalavathi	Hyderabad	Withdrawal	3500.00	2024-01-11
	106	Govinda	Vizag	Deposit	15000.00	2024-01-12
	107	Krishna	Vijayawada	Withdrawal	1000.00	2024-01-13
	108	Sairam	Hyderabad	Deposit	9000.00	2024-01-14
	109	Laxmi	Vizag	Withdrawal	4000.00	2024-01-15
	110	Saraswati	Vijayawada	Deposit	11000.00	2024-01-16
*	NULL	NULL	NULL	NULL	NULL	NULL

Insert

```
INSERT INTO bank_transactions
VALUES
(111,'Venu','Vizag','Deposit',7000,'2024-01-18');
SELECT * FROM bank_transactions;
```

Output:

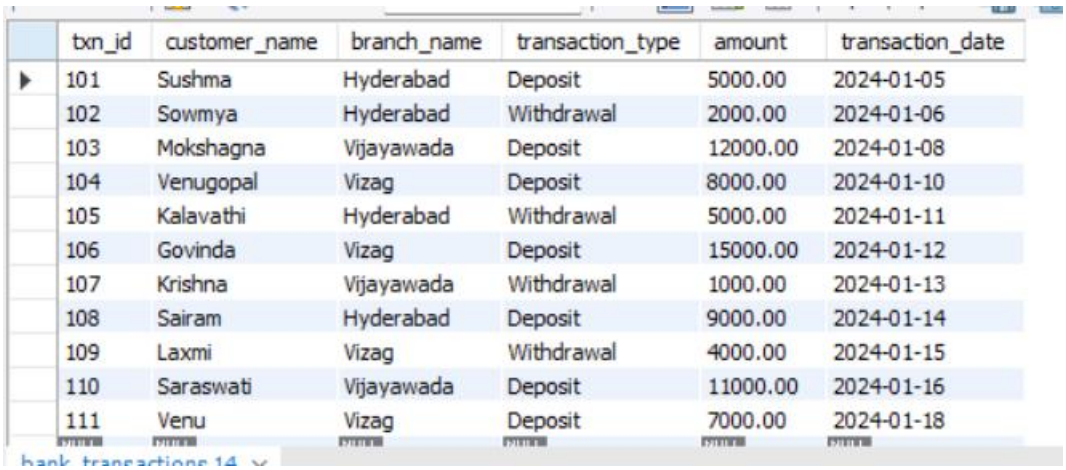


txn_id	customer_name	branch_name	transaction_type	amount	transaction_date
101	Sushma	Hyderabad	Deposit	5000.00	2024-01-05
102	Sowmya	Hyderabad	Withdrawal	2000.00	2024-01-06
103	Mokshagna	Vijayawada	Deposit	12000.00	2024-01-08
104	Venugopal	Vizag	Deposit	8000.00	2024-01-10
105	Kalavathi	Hyderabad	Withdrawal	5000.00	2024-01-11
106	Govinda	Vizag	Deposit	15000.00	2024-01-12
107	Krishna	Vijayawada	Withdrawal	1000.00	2024-01-13
108	Sairam	Hyderabad	Deposit	9000.00	2024-01-14
109	Laxmi	Vizag	Withdrawal	4000.00	2024-01-15
110	Saraswati	Vijayawada	Deposit	11000.00	2024-01-16
111	Venu	Vizag	Deposit	7000.00	2024-01-18

Update:

```
UPDATE bank_transactions
SET amount = 5000
WHERE txn_id = 105;
SELECT * FROM bank_transactions;
```

Output:



txn_id	customer_name	branch_name	transaction_type	amount	transaction_date
101	Sushma	Hyderabad	Deposit	5000.00	2024-01-05
102	Sowmya	Hyderabad	Withdrawal	2000.00	2024-01-06
103	Mokshagna	Vijayawada	Deposit	12000.00	2024-01-08
104	Venugopal	Vizag	Deposit	8000.00	2024-01-10
105	Kalavathi	Hyderabad	Withdrawal	5000.00	2024-01-11
106	Govinda	Vizag	Deposit	15000.00	2024-01-12
107	Krishna	Vijayawada	Withdrawal	1000.00	2024-01-13
108	Sairam	Hyderabad	Deposit	9000.00	2024-01-14
109	Laxmi	Vizag	Withdrawal	4000.00	2024-01-15
110	Saraswati	Vijayawada	Deposit	11000.00	2024-01-16
111	Venu	Vizag	Deposit	7000.00	2024-01-18

Delete & Select

```
DELETE FROM bank_transactions
WHERE txn_id = 111;
SELECT * FROM bank_transactions;
```

Output:

txn_id	customer_name	branch_name	transaction_type	amount	transaction_dat
101	Sushma	Hyderabad	Deposit	5000.00	2024-01-05
102	Sowmya	Hyderabad	Withdrawal	2000.00	2024-01-06
103	Mokshagna	Vijayawada	Deposit	12000.00	2024-01-08
104	Venugopal	Vizag	Deposit	8000.00	2024-01-10
105	Kalavathi	Hyderabad	Withdrawal	5000.00	2024-01-11
106	Govinda	Vizag	Deposit	15000.00	2024-01-12
107	Krishna	Vijayawada	Withdrawal	1000.00	2024-01-13
108	Sairam	Hyderabad	Deposit	9000.00	2024-01-14
109	Laxmi	Vizag	Withdrawal	4000.00	2024-01-15
110	Saraswati	Vijayawada	Deposit	11000.00	2024-01-16
NULL	NULL	NULL	NULL	NULL	NULL

Select with where

```
SELECT *
FROM bank_transactions
WHERE transaction_type='Deposit';
```

Output:

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date
101	Sushma	Hyderabad	Deposit	5000.00	2024-01-05
103	Mokshagna	Vijayawada	Deposit	12000.00	2024-01-08
104	Venugopal	Vizag	Deposit	8000.00	2024-01-10
106	Govinda	Vizag	Deposit	15000.00	2024-01-12
108	Sairam	Hyderabad	Deposit	9000.00	2024-01-14
110	Saraswati	Vijayawada	Deposit	11000.00	2024-01-16
NULL	NULL	NULL	NULL	NULL	NULL

Select with order by

```
SELECT *
FROM bank_transactions
ORDER BY amount DESC;
```

Output:

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date
106	Govinda	Vizag	Deposit	15000.00	2024-01-12
103	Mokshagna	Vijayawada	Deposit	12000.00	2024-01-08
110	Saraswati	Vijayawada	Deposit	11000.00	2024-01-16
108	Sairam	Hyderabad	Deposit	9000.00	2024-01-14
104	Venugopal	Vizag	Deposit	8000.00	2024-01-10
101	Sushma	Hyderabad	Deposit	5000.00	2024-01-05
105	Kalavathi	Hyderabad	Withdrawal	5000.00	2024-01-11
109	Laxmi	Vizag	Withdrawal	4000.00	2024-01-15
102	Sowmya	Hyderabad	Withdrawal	2000.00	2024-01-06
107	Krishna	Vijayawada	Withdrawal	1000.00	2024-01-13
NULL	NULL	NULL	NULL	NULL	NULL

DCL (DATA CONTROL LANGUAGE)

Grant all privileges

```
CREATE USER 'BranchManager'@'localhost'
IDENTIFIED BY 'Branch@123';
GRANT ALL PRIVILEGES
ON bookflow1_db.bank_transactions
TO 'BranchManager'@'localhost';
SHOW GRANTS FOR 'BranchManager'@'localhost';
```

Output:

```
97  22:36:21  SHOW GRANTS FOR 'BranchManager'@'localhost'
```

Revoke all

```
REVOKE ALL PRIVILEGES
ON bookflow1_db.bank_transactions
FROM 'BranchManager'@'localhost';
```

Output:

```
98  22:38:01  REVOKE ALL PRIVILEGES ON bookflow1_db.bank_transactions FROM 'BranchManager'@'localhost'
```

TCL (TRANSACTION CONTROL LANGUAGE)

Commit

```
UPDATE bank_transactions
SET amount=6000
WHERE txn_id=101;
COMMIT;
```

Output:

```
105  22:45:10  UPDATE bank_transactions SET amount=6000 WHERE txn_id=101
106  22:45:12  COMMIT
```

Savepoint:

```
SAVEPOINT Before_Update;
```

Output:

```
107  22:46:42  SAVEPOINT Before_Update
```

Rollback To Savepoint

```sql
UPDATE bank_transactions
SET amount=99999
WHERE txn_id=102;
```

```sql
SELECT * FROM bank_transactions;
```

Output:

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date
101	Sushma	Hyderabad	Deposit	6000.00	2024-01-05
102	Sowmya	Hyderabad	Withdrawal	99999.00	2024-01-06
103	Mokshagna	Vijayawada	Deposit	12000.00	2024-01-08
104	Venugopal	Vizag	Deposit	8000.00	2024-01-10
105	Kalavathi	Hyderabad	Withdrawal	5000.00	2024-01-11
106	Govinda	Vizag	Deposit	15000.00	2024-01-12
107	Krishna	Vijayawada	Withdrawal	1000.00	2024-01-13
108	Sairam	Hyderabad	Deposit	9000.00	2024-01-14
109	Laxmi	Vizag	Withdrawal	4000.00	2024-01-15
110	Saraswati	Vijayawada	Deposit	11000.00	2024-01-16
NULL	NULL	NULL	NULL	NULL	NULL